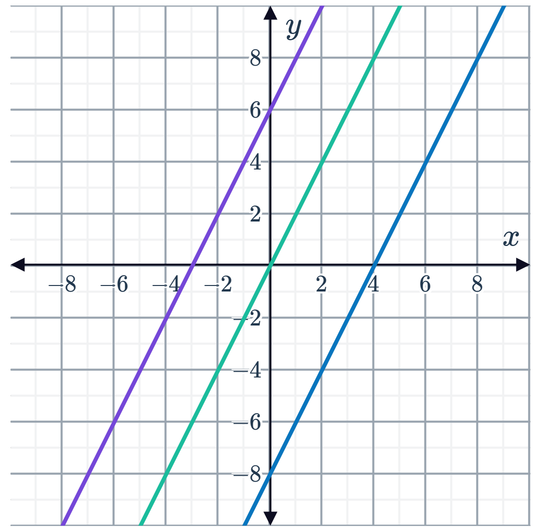
Equations of parallel and perpendicu[redacted]nes

1 The equations $y = 2x$, $y = 2x + 6$ and $y = 2x - 8$ have been graphed on the same number plane:



a Describe the three equations in terms of their slope.

b Determine whether the following statements describe the given graphs.

i All graphs cut the y-axis at the same point.

ii All graphs cut the x-axis at the same point.

iii All graphs have the same angle of inclination.

c What can you conclude from the answers above?

2 Determine whether the following pairs of lines are parallel.

a $y = 4x - 1$ and $y = 4x - 6$

b $y = -8x - 2$ and $y = 9x + 7$

c $y = -6x - 5$ and $y = x$

3 Determine whether the following lines are parallel to $y = 9x + 2$.

a $y = 9x$

b $y = 5x + 5$

c $y = 9x + 5$

d $y = -9x + 2$

e $y = 9x - 2$

f $y = 5x + 2$

4 Given that the line formed by Equation 1 is parallel to the line formed by Equation 2, find the missing value for each of the following.

a Equation 1: $y = \square x + 4$
Equation 2: $y = -3x - 7$

b Equation 1: $y = \square x + 4$
Equation 2: $y = \dfrac{4}{5}x - 7$

5 Find the two equations of the line that are both parallel to $y = -4$, and 2 units away from the line $y = -4$.

Determine the equation of the original line.

7 Find the equation of the following lines, expressing your answer in slope-intercept form when appropriate.

a A line that is parallel to the x-axis and passes through $(-8, 5)$

b A line that is parallel to the y-axis and passes through $(-7, 5)$

c A line that is parallel to $y = -\frac{7x}{6} - 4$ and passes through the point $(0, -2)$

d A line that is parallel to $y = -8x - 6$ and crosses the y-axis at $(0, 7)$

e A line that is parallel to $y = 2x - 1$ and passes through the point $(1, 3)$

f A line that is parallel to $y = \frac{2x}{3} - 10$ and passes through $(-3, -3)$

8 Find the equation of the line that is parallel to $y = \frac{4x}{3} - 3$ and passes through $(-2, -5)$. Express your answer in standard form.

9 A line passes through points $A\,(4, 3)$ and $B\,(8, -10)$.

a Find the slope of the line.

b Find the equation of a line that passes through $(-1, 2)$ and is parallel to the line passing through $A\,(4, 3)$ and $B\,(8, -10)$. Express the equation in slope-intercept form.

10 A line passes through $A\,(2, -4)$ and $B\,(3, 9)$.

a Find the slope of this line.

b Find the equation of a line that has a y-intercept of 5 and is parallel to the line that goes through $A\,(2, -4)$ and $B\,(3, 9)$. Express the equation in slope-intercept form.

11 The lines $y = -5mx + 1$ and $y = -2 + 4x$ are parallel.
Find the value of m.

12 For each of the following:

i Find the slope of the line.

ii Find the value of b if the equation is to be written in the form $y = mx + b$.

iii Write the equation of this line in slope-intercept or standard form.

a A line passes through the point $(9, 4)$ and is parallel to the line with equation $y = 4x + 2$

b A line passes through the point $(8, -8)$ and is parallel to the line with equation $y = -x + 5$

c A line passes through the point $(-6, -4)$ and is parallel to the line with equation $y = 5x - 4$

13 Find the negative reciprocal of the following:

a $\frac{5}{2}$

b $-\frac{10}{9}$

c 3

d $-\frac{1}{4}$

14 Determine which is the single best definition that describes two perpendicular lines.

- Two lines that have no intersections.
- Two lines that have only one intersection.
- Two lines whose product of slopes is equal to 1.
- Two lines that intersect each other at an angle of 90°.

15 Given that two non-vertical lines are perpendicular, state the product of their slopes.

16 State whether the following pairs of lines are perpendicular.

a $y = x + 6$ and $y = x - 6$

b $y = 6x - 1$ and $y = -6x - 4$

c $y = 4x - 8$ and $y = -\frac{x}{4} + 3$

d $y = \frac{5x}{4} + 5$ and $y = 9 - \frac{4x}{5}$

e $y = \frac{x}{3} - 7$ and $y = -(3x + 21)$

f $y = 7x - 5$ and $y = -7x + 6$

g $y = -8x - 3$ and $y = \frac{x}{8} - 2$

17 Find the slope of the following lines.

a A line perpendicular to a line which has a slope of -9

b A line perpendicular to $y = -\frac{x}{3} - 7$

c Any line perpendicular to $y = -5x + 6$

d Any line that is perpendicular to $y = 10 - x$

18 Given that the line formed by Equation 1 is perpendicular to the line formed by Equation 2, find the missing value for each of the following.

a Equation 1: $y = 10x - 6$
Equation 2: $y = \square x + 7$

b Equation 1: $y = \frac{3}{4}x + 2$
Equation 2: $y = \square x + 1$

c Equation 1: $y = -10x + 4$
Equation 2: $y = \square x - 6$

d Equation 1: $y = \frac{2}{5}x - 10$
Equation 2: $y = \square x - 5$

19 Determine whether the following lines are perpendicular to $y = 4x - 3$.

a $y - \frac{x}{4} = \frac{3}{2}$

b $4x + 16y = 96$

c $x + 4y = -3$

d $y = -4x - 3$

e $y = \frac{x}{4} - 3$

20 Determine whether the following is an equation of the line that is perpendicular to $3y = 8 + \frac{3x}{4}$ and passes through $(-4, 8)$.

a $y = \frac{x}{4} + 9$

b $y = \frac{x}{4} - 6$

c $y = -4x + 28$

d $y = -4x - 8$

21 Write down the equation of the following lines, expressing your answers in standard form.

a A line that is perpendicular to the x-axis and passes through $(10, 3)$

b A line that is perpendicular to the y-axis and passes through $(1, -7)$

c A line that is perpendicular to $y = 7x - 7$ and passes through $(-1, 1)$

d A line that is perpendicular to $y = \frac{7x}{4} + 9$ and passes through $(4, -4)$

e A line that is perpendicular to $2x - 7y = -5$, and passes through $(-5, 2)$

22 A line L is perpendicular to the segment joi[illegible]$B\,(2,-10)$ and $C\,(-1,10)$.

- **a** Find the slope of line L.
- **b** Find the equation of line L if it passes through $(-8,-4)$. Express the equation in slope-intercept or standard form.

23 Consider the line $4x-3y=-6$.

- **a** Find the y-intercept of the line.
- **b** Find the equation of the line that is perpendicular to the given line and has the same y-intercept. Express the equation in standard form.

24 Consider the line that is perpendicular to $3x-8y=-2$ and passes through $(1,2)$.

- **a** Find the equation of the line in standard form.
- **b** Does the point $\left(-3,\dfrac{38}{3}\right)$ lie on the line?

25 Consider the line that is perpendicular to $5x+2y=-7$ and passes through $(4,0)$.

- **a** Find the equation of the line in standard form.
- **b** Does the point $\left(-4,-\dfrac{11}{5}\right)$ lie on the line?

26 The line L_1 is perpendicular to $y=5x+6$ and intersects the y-axis at 6.

- **a** Determine the slope of L_1.
- **b** State the equation of L_1, writing your answer in slope-intercept form.

27 Find the equation of a line that is perpendicular to $y=10x+3$, and has the same y-intercept. Express the equations in slope-intercept or standard form.

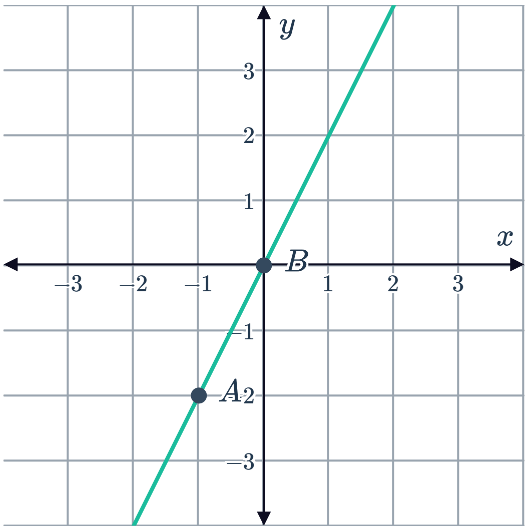
Additional questions

28 For each of the following:

i Write the slope of a line that would be parallel to the graphed line.

ii Write the slope of a line that would be perpendicular to the graphed line.

a



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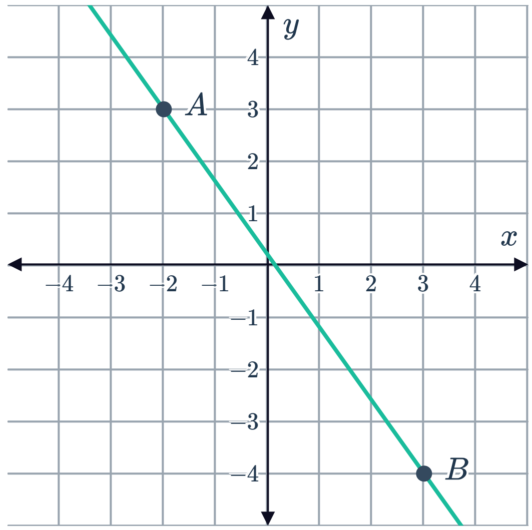

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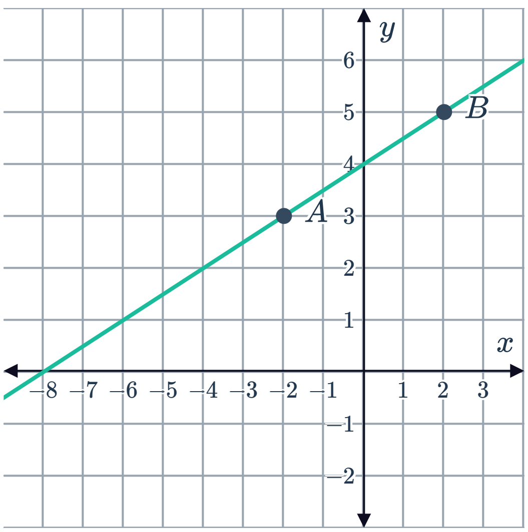

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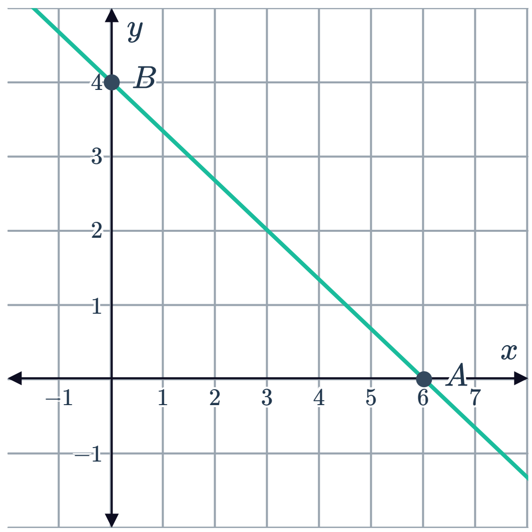
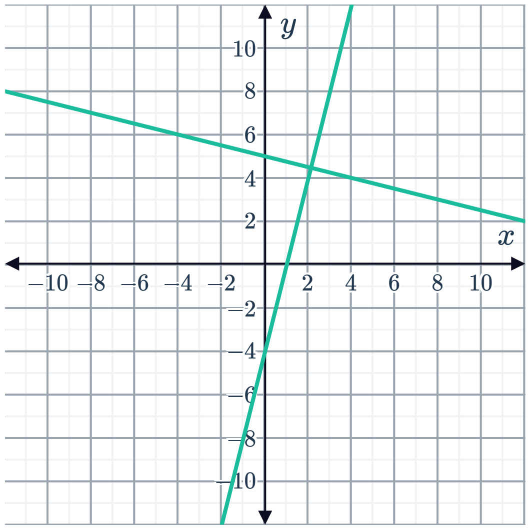


29 Determine whether the following lines are p[redacted]el, perpendicular, or neither.

a



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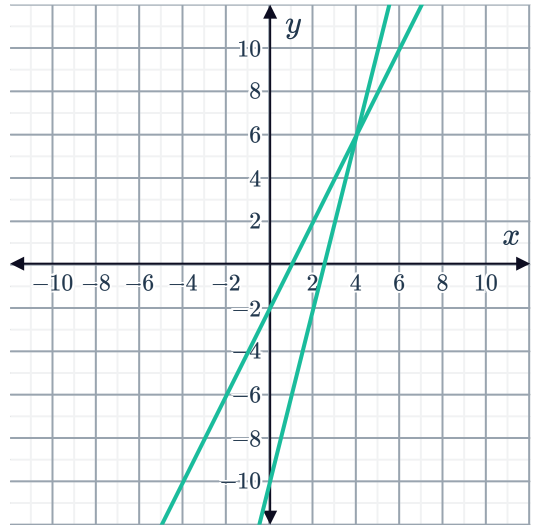

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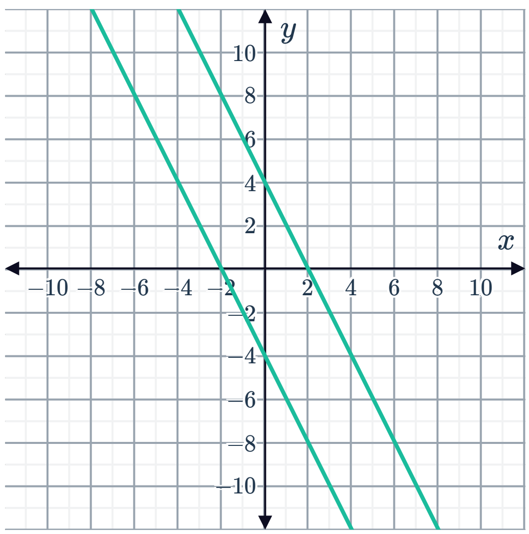

30 State the relationship between the slopes of

a Parallel lines.

b Perpendicular lines.

31 Find the slope, m, of a line that is

a Parallel to a line which has a slope of 4.

b Parallel to $y = -\dfrac{x}{3} - 7$.

c Perpendicular to a line which has a slope of -9.

d Perpendicular to $y = -5x + 6$.

32 Determine if the following lines are parallel to $y = \frac{2}{3}x - 3$ or not.

a $y = \frac{2}{3}x + 1$

b $4x - 6y = 37$

c $3x + 2y = -13$

d $y = -\frac{2}{3}x - 8$

33 Determine if the following lines are perpendicular to $y = 4x - 3$ or not.

a $y = \frac{x}{4} - \frac{3}{2}$

b $4x + 16y = 96$

c $x + 4y = -3$

d $y = -4x - 3$

34 For each of the following pair of lines, determine whether they are parallel, perpendicular, or neither.

a $y = 3x - 7$ and $y = 3x + 4$

b $x - y = 6$ and $x - y = -6$.

c $8x + y = 2$ and $y = \frac{x}{8} - 5$

d $2x + 3y = 5$ and $-3x - 2y = 12$

35 Find the equation the line that is

a Parallel to the line $y = -\frac{7x}{6} - 4$ and passes through the point $(0, -2)$.

b Parallel to $x = 1$ and passes through $(-4, 6)$.

c Parallel to the x-axis and passes through $(-8, 5)$.

d Parallel to the y-axis and passes through $(-7, 5)$.

36 Find the equation the line that is

a Perpendicular to $x = 4$ and passes through $(-7, 1)$.

b Perpendicular to $y = 2x - 1$ and passes through the point $(2, 3)$.

c Perpendicular to the x-axis and passes through the point $(10, 3)$.

d Perpendicular to the y-axis and passes through $(1, -7)$.

37 Consider a line that passes through points $A\,(4, 3)$ and $B\,(8, -10)$.

a Find the slope, m, of the line.

b Find the equation of a line that passes through $(-1, 2)$ and is parallel to the line passing through $A\,(4, 3)$ and $B\,(8, -10)$. Express the equation in slope-intercept form.

38 Consider a line that passes through $A\,(2,-4)$ and $B\,(3,9)$.

- **a** Find the slope, m, of the line.
- **b** Find the equation of a line that has a y-intercept of 5 and is perpendicular to the line that goes through $A\,(2,-4)$ and $B\,(3,9)$. Express the equation in slope-intercept form.

39 Consider line L_1 with equation $3x-4y-4=0$.

- **a** Write L_1 in slope-intercept form.
- **b** Find the slope of a line, L_2, that is perpendicular to L_1.
- **c** Find the equation of L_2 in standard form, given that it passes through the point $A\,(-6,4)$.

40 Consider the lines $2x-7y=2$ and $mx-35y=-7$.
Given that the two lines are parallel, determine the value of m.

41 Explain why two lines with the standard forms $ax+by=c_1$ and $bx-ay=c_2$ are perpendicular.

42 Ammon and Minh work part-time at the same pizzeria, both earning $\$360$ per week. Both Ammon and Minh are saving up, so they don't spend any of the money they earn. Ammon currently has a bank balance of $\$212$ while Minh has $\$275$.
If we sketch linear graphs for Ammon and Minh's bank balances over time, can we expect to get two parallel lines? Justify your answer.

43 Suppose that the lines $px-4y=7$ and $qx+5y=-14$ are perpendicular.
Write an equation relating the values of p and q.